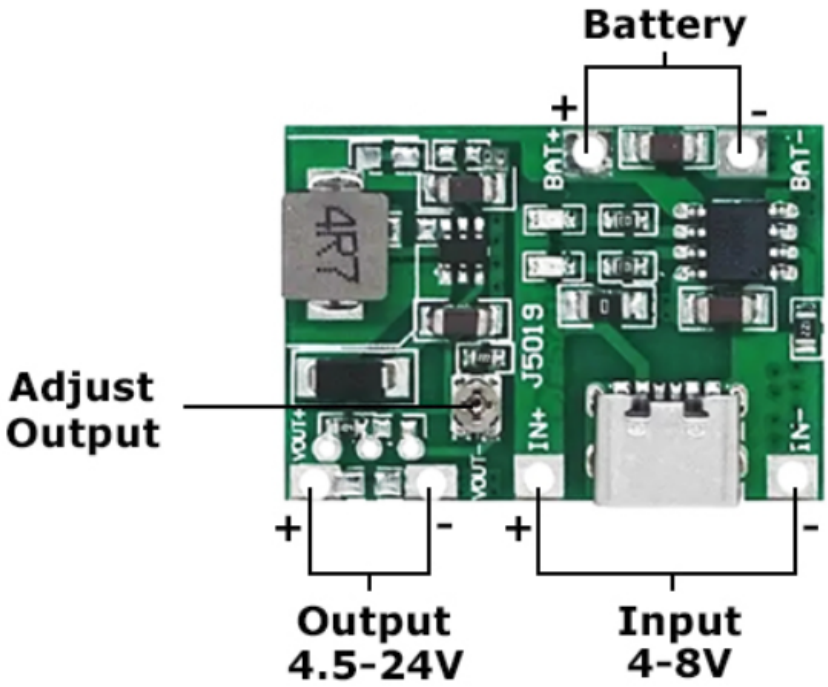
Quick Spec

- Input Voltage (USB Type-C): 5V
- Input Voltage (IN): 4 - 8V
- Battery Type: LiPo 3.7V
- Default Charging Current: 1000mA
- Output Voltage: 4.5 - 24V
- Max Output Current: 1A
- Dimensions: 33 x 24mm
- Weight: 6g

Board Layout

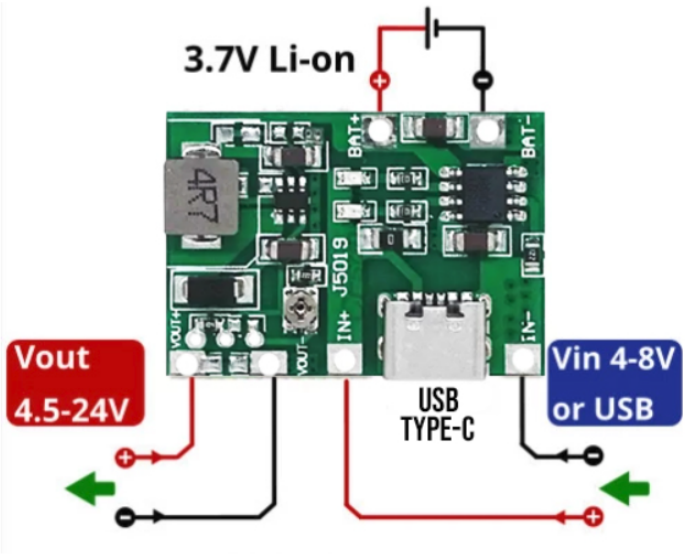


Shipping List

- 1 x LiPo Charger 3.7V with DC-DC Boost 4.5-24V - Type C

LiPo Charger 3.7V with DC-DC Boost 4.5-24V - Type C

This module is a small single cell lithium battery charging module which also includes a 1A step-up (boost) converter for powering a large range of applications. The module will charge most types of single cell (3.7) LiPo batteries from either 4 to 8V power supply input, or from a standard 5V USB Type-C port/adapter. A battery charge and standby LED is also included for visual indication.



Besides battery charging capabilities this module also includes an adjustable boost converter which is capable of stepping up the attached battery voltage from 4.5 to 24V with a maximum supply current of 1A.

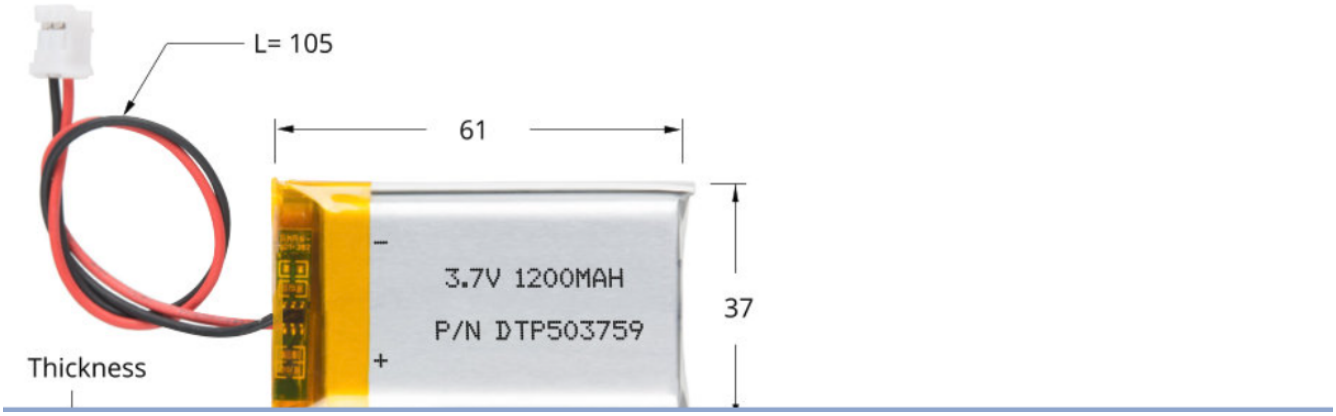
Battery LiPo 1200mAh 3.7V

Lipo Batteries (Polymer Lithium Ion) offers extremely light weight and high energy density. This battery is rated at 3.7V 1200mAh and is terminated with a 2-pin JST-PH connector (2mm pin spacing). These batteries require special charger so please do not attempt to charge these with anything but a specialized Lithium Polymer charger.

Quick Spec

- Capacity: 1200mAh
- Nominal Voltage: 3.7V
- Open Circuit Voltage: 3.75 to 4.2V
- Cut Off Voltage: 2.4V
- Standard Charge Current: 240mA
- Max Continuous Charge Current: 1C (1200mA)
- Dimensions: 61mm x 37mm x 5.1mm
- Protection Circuit: Yes (DW01 IC)
- Connector: JST-PH
- Weight: 24g

Dimensions



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Backup Battery Switch Over Module

The XH-M350 module facilitates an automatic switchover from the main power source to a backup when the primary source fails. It incorporates a high-quality industrial relay with an effective control current capacity of up to 5A.

Quick Spec

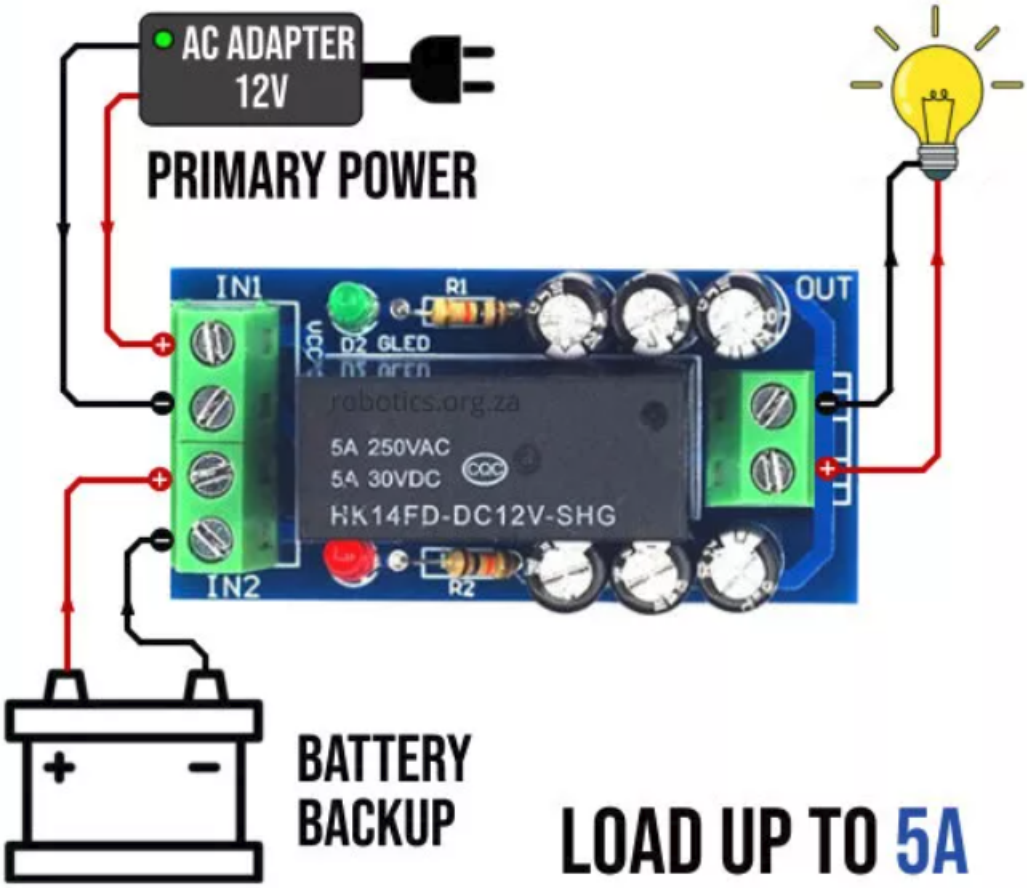
- Model: XH-M350
- Control voltage: 12VDC
- Control current: 5A
- Support power: 60W
- Size: 54*26mm

Working Principle Diagram

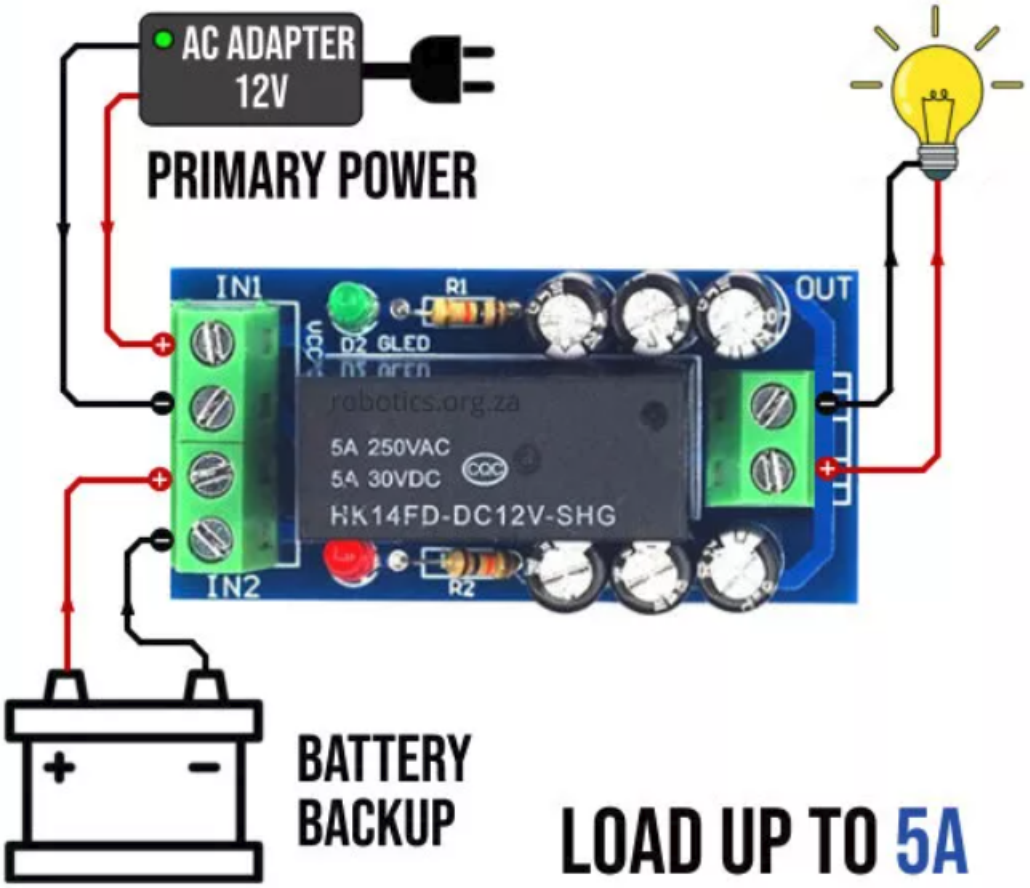
- Connect the 12V Adapter as shown in the diagram.
- When the Grid power is connected the Green Led is On
- When Grid power Fails, the unit switch automatically over to Backup Battery and RED is On

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AUTOMATIC BATTERY SWITCH OVER



LOAD UP TO 5A

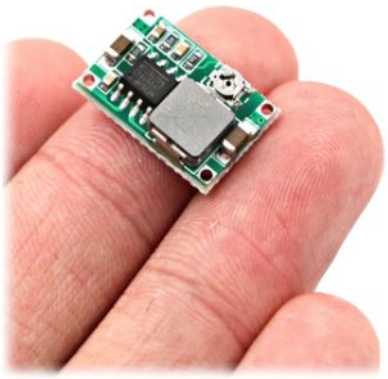


LOAD UP TO 5A

LM2596 Mini Buck Regulator VIN 5-18V OUT 1-17V 2A

The mini 360 step down converter is a monolithic synchronous reversing electronic device, which allows converting direct current “DC” voltages to another “DC” voltage with high conversion efficiency. This device has an integrated MP1482 chip that integrates 130mΩ MOSFETS to provide 2A of continuous load current over a wide input voltage and operation of 4.75V to 18V.

This mini voltage converter is perfect to apply it in your electronics projects where you need to maintain the power you want to supply, due to its small size and easy connection, it is considered a very versatile and practical device when using it.



This device supplies an output voltage lower than the input voltage, it can operate with a voltage from 4.75 to 18 V, at a switching frequency of 340 KHz and a load current of 1.8 A continuously and up to 2 A during a short time.

Finally, it has a variable potentiometer that allows the required output voltage to be adjusted in steps of ± 2.5%, and the maximum difference between the output and input voltage should not exceed 15V. There is no protection against reverse polarity, but against overheating which shuts down the module when it reaches 160°C.

Quick Spec

- Model: MH-MINI-360
- Input Voltage: 5 ~ 18V
- Output Voltage: 1 ~ 17V
- Output Current: 0-2A
- Conversion Efficiency: 93%
- Switching Frequency: 340kHz
- Output Ripple: 30mV (No load)
- Load Regulation: ±0.5%
- Voltage Regulation: ±2.5%
- Operating Temperature: -40 ~ +85°C
- Dimensions: 18 x 12 x 4.5mm
- Weight: 3g